

A Theory of Economic Slack

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APPENDIX E.

Alternative matching cost

We have seen in chapter 5 that assessing the matching cost in terms of goods makes sense for markets in which the goods are labor or services. It is less clear how the assumption applies to markets for actual goods. In this appendix, we introduce the matching cost in a different way: as the cost of a sample that the buyer procures to determine whether they want to purchase the good or not. We show that this alternative representation is equivalent to the representation used in the model, and that it applies to many goods markets.

E.1. Model with sampling

Consider a market as described in chapter 5. Each of the ν visits brings a buyer to one of the k goods offered for sale.

The buyer does not know whether they would like the good or not, having never bought it before. Hence the buyer buys a sample of the good to determine whether they want to purchase the good or not. The sample has a small size κ and is sold at price κp .

Samples do not contribute to consumption: they are used only for tasting. Only goods purchased after tasting contribute to consumption.

The probability that the buyer is happy with the good and purchases the full unit is set to $q(\theta) - \kappa$, where $q(\theta) = m(1/\theta, 1)$, as usual. In this interpretation, $q(\theta)$ is no longer the probability of purchasing the full-unit good. Instead, it represents the expected total quantity of goods purchased per visit: sample plus possible full-unit purchase. Note that

$q(\theta) \leq 1$, and $q(\theta) \geq \kappa$ because $\theta \leq \bar{\theta}$, so $q(\theta) - \kappa$ is a proper probability: it is between 0 and 1. The probability that the buyer does not ultimately buy the good—because they do not like it—is $1 - [q(\theta) - \kappa]$. Because $q(\theta) - \kappa$ is a proper probability, so is $1 - [q(\theta) - \kappa]$.

We parameterize the probability of purchasing the full unit as $q(\theta) - \kappa$ so that, together with the sure purchase of a sample of size κ , the expected total quantity purchased per visit remains equal to $q(\theta)$, and the matching function continues to give the total quantity of goods traded—including samples and full-unit goods.

E.2. Matching wedge with sampling

The market with sampling behaves exactly like the baseline market described in chapter 5.

Buyers make v visits. Each visit costs κ goods. The total amount of goods purchased is

$$y = v\kappa + v[q(\theta) - \kappa] = vq(\theta) = m(k, v),$$

since $q(\theta) = m(k/v, 1)$ and the matching function has constant returns to scale. The total amount of goods consumed simply is

$$c = y - v\kappa.$$

The numbers of goods purchased and sold are equal, so $m(k, v)$ goods are sold in total. This means that seller i , who places a share k_i/k of all goods for sale, sells an amount of goods given by

$$m(k, v) \cdot \frac{k_i}{k} = m(1, \theta) \cdot k_i = f(\theta)k_i.$$

On average, it is as if each good is sold with probability $f(\theta)$.

These sales take two forms: some goods are sold in the form of samples, and some in the form of complete goods to be consumed. Seller i receives a number $v \cdot k_i/k = \theta k_i$ of the visits, so sells $\theta k_i \kappa$ samples, and in addition sells $\theta k_i [q(\theta) - \kappa]$ complete goods. The total amount of sales therefore is $\theta k_i q(\theta) = f(\theta)k_i$, since $\theta q(\theta) = f(\theta)$.

The wedge between consumption and sales is

$$\tau(\theta) = \frac{y - c}{c} = \frac{\kappa v}{m(k, v) - \kappa v} = \frac{\kappa}{m(k, v)/v - \kappa} = \frac{\kappa}{q(\theta) - \kappa},$$

using the fact that the matching function has constant returns to scale. This is just the expression for the matching wedge in the baseline model, given by (5.8).

All in all, the market operates exactly as in the baseline model.

E.3. Markets with sampling in the real world

Many goods markets can be described using this equivalent representation. The wine market is a perfect example: people go to wineries, buy glasses of wine to taste different wines, and purchase bottles of the wines they enjoy.

The online market for goods also fits the sample representation, once the price paid for a good is separated between shipping and restocking fees and the rest. For many goods bought online, there is no guarantee that the good fits the buyer's needs. Indeed it's very hard to tell without seeing the good and trying it: most goods are experience goods (Che 1996). Goods that buyers do not like can be returned, but not for free: the buyer is typically not reimbursed for shipping and is commonly assessed a restocking fee (Janssen and Williams 2024). These fees, the logistical effort to bring the good back and repack it, and the associated depreciation of the good, correspond to the sample sold by the seller. Buyers who keep their online purchases are the ones who actually consume the goods.

Any good sold in brick-and-mortar stores charging a restocking fee could also be described as above.

As a final example, many goods are sold in stores that offer the good both for short-term rental and for sale. Buyers can rent the good to try it out, and then buy it if they like it. Many surf shops and ski shops operate that way. Renting the good to test it corresponds to buying a sample; purchasing the good corresponds to consuming it.

Even internships on the labor market fit here: hiring an intern at a reduced wage for a short time corresponds to "sampling the good," and employing a well-performing intern at full pay on a long-term contract corresponds to "purchasing the good."

Bibliography

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